

JEE MAIN REVISION MASTER: ELECTROMAGNETIC INDUCTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 21 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Magnetic Flux & Faraday-Lenz Law	14 PYQs	Core Principle
Moderate	Motional EMF (Translational & Rotational)	4 PYQs	Motional Induction
Moderate	Self Inductance & RL Transient Circuits	4 PYQs	Inductance & Circuits
Standard	Mutual Inductance & Coupled Coils	2 PYQs	Coupling
Standard	Induced Electric Field & Eddy Currents	1 PYQs	Field Theory

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Magnetic Flux & Faraday-Lenz Law	$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos\theta$, $\mathcal{E} = -\frac{d\Phi_B}{dt}$, $\mathcal{E} = -N \frac{d\Phi_B}{dt}$, $q = \frac{\Delta \Phi}{R}$
Motional EMF (Translational & Rotational)	$\mathcal{E} = B v L$, $I = \frac{B v L}{R}$, $F_{\text{mag}} = \frac{B^2 L^2 v}{R}$, $\mathcal{E}_{\text{rot}} = \frac{1}{2} B \omega L^2$
Self Inductance & RL Transient Circuits	$\mathcal{E} = -L \frac{dI}{dt}$, $L = \mu_0 n^2 A l$, $U = \frac{1}{2} L I^2$, $\tau = \frac{L}{R}$, $I(t) = \frac{V}{R} (1 - e^{-t/\tau})$
Mutual Inductance & Coupled Coils	$\mathcal{E}_2 = -M \frac{dI_1}{dt}$, $M = \mu_0 n_1 n_2 (\pi r_{\text{inner}}^2) l$, $M = k \sqrt{L_1 L_2}$ ($0 \leq k \leq 1$)
Induced Electric Field & Eddy Currents	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$, $E(r) = \frac{r}{2} \frac{dB}{dt}$ ($r \leq R$), $E(r) = \frac{R^2}{2r} \frac{dB}{dt}$ ($r > R$)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Flux & Faraday-Lenz Law (High Yield — 14 PYQs)

Core Principle: Magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = B A \cos\theta$. Induced EMF $\mathcal{E} = -(\mathrm{d}\Phi_B / \mathrm{d}t) = -N (\mathrm{d}\Phi_B / \mathrm{d}t)$. Lenz's law dictates that induced currents oppose the change in magnetic flux.

■ **Caution / JEE Pitfall:** 1. Total induced charge flown $\Delta q = (\Delta\Phi_B / R)$ is independent of the rate or time interval of flux change! 2. Angle θ is measured from the surface normal $\vec{A}$, not the coil plane.

■ **Speed Trick:** When an N -turn coil rotates with angular velocity ω in field B : $\mathcal{E}(t) = N B A \omega \sin(\omega t)$, with peak EMF $\mathcal{E}_0 = N B A \omega$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In the figure shown, the conductor PQ of length l is moved from $x=0$ to $x=b$ and then up to $x=2b$ with a constant velocity $\vec{v}$ " **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "In Faraday-Henry's experiment, a coil is connected to a galvanometer. For the deflection of pointer in the galvanometer, which of the follow..." **[Ans: N/A]**
- **[2025 Shift] Q#3 (MCQ):** "When a bar magnet is pushed towards the coil, along its axis, as shown in the figure, the galvanometer pointer deflects towards X . When thi..." **[Ans: N/A]**

3.2 Motional EMF (Translational & Rotational) (Moderate Yield — 4 PYQs)

Core Principle: Motional EMF in a conductor moving through field $\vec{B}$: $\mathcal{E} = B v L$. For a rod rotating about one pivot with angular speed ω : $\mathcal{E}_{\text{rot}} = (1/2) B \omega L^2$.

■ **Caution / JEE Pitfall:** 1. Mechanical power delivered by external pulling force $P_{\text{mech}} = F_{\text{ext}} v$ exactly matches electrical Joule dissipation $I^2 R = (B^2 L^2 v^2 / R)$. 2. For an arbitrary curved wire, replace it with the straight vector connecting its endpoints.

■ **Speed Trick:** Direction of induced EMF follows $\vec{v} \times \vec{B}$. Positive charges accumulate at the end pointed to by $\vec{v} \times \vec{B}$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In the figure shown, the conductor PQ of length l is moved from $x=0$ to $x=b$ and then up to $x=2b$ with a constant velocity $\vec{v}$ " **[Ans: N/A]**
- **[2023 Shift] Q#1 (MCQ):** "A metallic rod of length 1m held along east-west direction is allowed to fall down freely. Given horizontal component of earth's magnetic ..." **[Ans: N/A]**
- **[2020 Shift] Q#2 (MCQ):** "A rod of length 2m slides with a speed of 5ms^{-1} on a rectangular conducting frame as shown in figure. There exists a uniform magnetic..." **[Ans: N/A]**

3.3 Self Inductance & RL Transient Circuits (Moderate Yield — 4 PYQs)

Core Principle: Self-inductance $L = (N\Phi / I)$, induced back-EMF $\mathcal{E} = -L(dI / dt)$. Solenoid $L = \mu_0 n^2 A l$. RL circuit current growth: $I(t) = (V / R)(1 - e^{-t/\tau})$ with inductive time constant $\tau = (L / R)$.

■ **Caution / JEE Pitfall:** At $t = 0^+$ (immediately after switch closing), an unenergized inductor acts as an open circuit ($R \rightarrow \infty$). At $t \rightarrow \infty$ (steady state), an ideal inductor acts as a zero-resistance short wire.

■ **Speed Trick:** Magnetic energy stored: $U_B = (1/2) L I^2$; energy density in space: $u_B = (B^2 / 2\mu_0)$. Inductors resist sudden changes in current, not voltage.

Representative Source PYQs:

- **[2023 Shift] Q#4 (MCQ):** "The current in a coil changes from 2A to 5A in 0.3s. The magnitude of emf induced in the coil is 1.0V. The value of self-inductance..." **[Ans: N/A]**
- **[2022 Shift] Q#2 (MCQ):** "A long solenoid has 500 turns, when a current of 2A is passed through it, the resulting magnetic flux linked with each turn of the solenoid..." **[Ans: N/A]**
- **[2021 Shift] Q#1 (MCQ):** "The current following through an inductance coil of self-inductance 6mH at different time instants is as shown. The emf induced between t_1 and t_2 ..." **[Ans: N/A]**

3.4 Mutual Inductance & Coupled Coils (Standard Yield — 2 PYQs)

Core Principle: Coupled flux in secondary coil $\Phi_2 = M I_1$, induced EMF $\mathcal{E}_2 = -M(dI_1 / dt)$. Coaxial solenoids $M = \mu_0 n_1 n_2 (\pi r_{\text{inner}}^2) l$. Coupling coefficient $M = k\sqrt{L_1 L_2}$ ($0 \leq k \leq 1$).

■ **Caution / JEE Pitfall:** For concentric coaxial coils, always use the cross-sectional area of the INNER coil (r_{inner}), because the magnetic field of the outer coil only penetrates the inner core.

■ **Speed Trick:** Reciprocity theorem holds: $M_{12} = M_{21}$ regardless of differing geometry or number of turns.

Representative Source PYQs:

- **[2024 Shift] Q#2 (MCQ):** "A square loop of side length a is moving away from an infinitely long current carrying conductor at a constant speed v as shown. Let x ..." **[Ans: C]**
- **[2021 Shift] Q#4 (MCQ):** "The physical quantity which is measured in the unit of wb A^{-1} is..." **[Ans: N/A]**

3.5 Induced Electric Field & Eddy Currents (Standard Yield — 1 PYQs)

Core Principle: A time-varying magnetic field generates a non-conservative, circular electric field: $\oint \vec{E} \cdot d\vec{l} = -(d\Phi_B / dt)$. Inside cylinder ($r \leq R$): $E(r) = (r/2)(dB / dt)$; outside ($r > R$): $E(r) = (R^2 / 2r)(dB / dt)$.

■ **Caution / JEE Pitfall:** Induced electric fields are non-electrostatic: field lines form continuous closed loops and work done along a closed circuit is non-zero ($\oint \vec{E} \cdot d\vec{l} = \mathcal{E} \neq 0$)!

■ **Speed Trick:** Eddy currents and hysteresis losses in transformer cores are minimized by using thin laminated sheets of silicon steel insulated by varnish.

Representative Source PYQs:

- **[2017 Shift] Q#3 (MCQ):** "The working of magnetic braking of trains is based on..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure shown, the conductor PQ of length l is moved from $x=0$ to $x=b$ and then up to $x=2b$ with a constant velocity $\vec{v}(v)$. A uniform magnetic field $\vec{B}$ is perpendicular to the plane of the paper and extends from $x=0$ to $x=b$ and it is zero from $x>b$. The magnitude of emf induced in the conductor is

- (A) $Blv; 0 \leq x \leq b$
- (B) Zero; $0 \leq x \leq b$
- (C) $Blv; 0 \leq x \leq b$
- (D) $Blv; b \leq x \leq 2b$

Correct Answer: None

Solution Analysis: When conductor PQ moves from $x=0$ to $x=b$, it is inside the uniform magnetic field $\vec{B}$. The magnetic field is perpendicular to the plane of the paper. The conductor has length l and moves with constant velocity $\vec{v}(v)$. For a straight conductor moving perpendicular to a magnetic field, the motional emf induced across its ends is $E=Blv$. Here, B , l , and v are constant. So, the induced emf remains constant while the conductor is in the region $0 \leq x \leq b$. When the conductor moves from $x=b$ to $x=2b$, it is outside the magnetic field region. In this region, the magne

Benchmark Problem 2: KCET 2023 (Q#1)

A metallic rod of length 1m held along east-west direction is allowed to fall down freely. Given horizontal component of earth's magnetic field $B_H = 3 \times 10^{-5} \text{ T}$. the emf induced in the rod at an instant $t=2\text{s}$ after it is released is (Take, $g=10 \text{ ms}^{-2}$)

- (A) $6 \times 10^{-4} \text{ V}$
- (B) $3 \times 10^{-3} \text{ V}$
- (C) $3 \times 10^{-4} \text{ V}$
- (D) $6 \times 10^{-3} \text{ V}$

Correct Answer: None

Solution Analysis: When the rod falls vertically downward, it cuts the horizontal component of Earth's magnetic field. The motional emf induced in a rod is $E=Blv$. Here, $B=B_H = 3 \times 10^{-5} \text{ T}$, $l=1\text{m}$, v is the speed of the rod after $t=2\text{s}$. Since the rod is falling freely from rest, $v=gt=10 \times 2=20 \text{ m s}^{-1}$. Now put the values in the formula: $E=3 \times 10^{-5} \times 1 \times 20$, $E=6 \times 10^{-4} \text{ V}$. So the induced emf is $6 \times 10^{-4} \text{ V}$. Option A.

Benchmark Problem 3: KCET 2023 (Q#4)

The current in a coil changes from 2A to 5A in 0.3s. The magnitude of emf induced in the coil is 1.0V. The value of self-inductance of the coil is

- (A) 1.0mH
- (B) 100mH
- (C) 0.1mH
- (D) 10mH

Correct Answer: None

Solution Analysis: $\Delta I = 5-2=3\text{A}$, $\Delta t = 0.3\text{s}$. Induced emf, $e=1\text{V}$. We know that, $e = (L \Delta I / \Delta t) \Rightarrow L = (e / (\Delta I / \Delta t)) = (1 / (3 / 0.3)) = 0.1\text{H} = 100\text{mH}$

Benchmark Problem 4: KCET 2017 (Q#3)

The working of magnetic braking of trains is based on

- (A) eddy current
- (B) pulsating current
- (C) alternating current
- (D) steady current

Correct Answer: None

Solution Analysis: Magnetic braking in trains works on the principle of eddy currents. Here's a quick breakdown: When a magnet moves relative to a conductive material (like aluminum), it induces circular currents (eddy currents) in that conductor. According to Lenz's law, these eddy currents create their own magnetic fields that oppose the motion causing them. This opposition produces a braking force, which helps in slowing down the train without physical friction. So, the correct answer is: Option A: eddy current.